ITE4116M FINAL YEAR PROJECT

PROJECT SUGGESTION FORM

Academic Year 2026/27

Submission Guide: Students must form a group of 3 to 4 members, discuss the project topic with your preferred supervisor, and obtain verbal agreement. Once agreed, the group must complete this Project Suggestion Document and upload the document file (**in both PDF & DOCX**) on Moodle during the FYP session by **14-SEP-2026.**

1. General Project Information

Proposed Project Title	AI and IoT Based Food Replenishment and Waste Reduction System
Assigned Project Supervisor	Macro (proposed; verbal agreement pending)
Programme / Class Code	Higher Diploma in AI and Smart Technology (IT114124) / Class AIST-2B
Student Group Number	Group 2

Student Member Details (Normally 3 to 4 members):

S/N	Student Name	Student ID	Email Address	Proposed Major Role / Area of Responsibility
1	Yuan Yuchen	250342840	250342840@stu.vtc.edu.hk	**Project Leader / Backend AI Lead (40%)** Requirements, APIs, food recognition, demand forecasting, replenishment and integration tests.
2	Xu Ziqian	250137555	250137555@stu.vtc.edu.hk	**IoT Hardware / Edge Processing (30%)** Sensor integration, Pi acquisition, calibration, filtering, tray events and hardware tests.
3	Li Wenhua	250329909	250329909@stu.vtc.edu.hk	**Web Application / UX Developer (30%)** Staff dashboard, QR menu and orders, live updates, bilingual UI and usability tests.

2. Problem Statement & Project Objectives

Instructions: Describe the business or technical problem in information technology that this project intends to address. Define 2-3 specific project objectives (what your solution intends to achieve for the target users/businesses).

Problem Statement:
Buffet and batch-prepared self-service outlets need timely replenishment without excess food at closing. Visual checks can miss low stock or batches already being prepared, leading to waiting, duplicate work and leftovers. Customer menus can also show outdated availability.
We propose continuous monitoring while trays remain in service, linking measurements, demand estimates and kitchen progress. Metafoodx and Controliza are commercial references; the prototype will be evaluated against simple replenishment baselines.

Project Objectives:
1. Implement continuous monitoring for up to three serving positions, recording food quantity, refills and surface temperature and flagging unreliable data.
2. Develop AI-assisted replenishment recommendations using demand, preparation lead time and work in progress; compare empty-tray time and leftovers against fixed-threshold and consumption-rate baselines.
3. Integrate bilingual staff tools and a supplementary QR menu/order interface; evaluate consistency, interruption recovery and usability.

3. Proposed Project Scope & Key Features

Instructions: Define the scope of your system. List and briefly describe the key modules, functions, or features that your group plans to implement. Projects must be implementation-oriented and student-centered.

Key Subsystems / Features:
A. Continuous tray monitoring. Monitor weight, visual dish identity and surface temperature throughout service; handle tare, tray movement, refills, occlusion and stale data.
B. Replenishment and kitchen tasks. Suggest timing and batch quantities using demand, preparation/closing times and work in progress. Staff confirm or adjust tasks and record completed refills.
C. Operations dashboard and evaluation. Show alerts and export empty-tray time, refill history, leftovers and disposal/reuse records.
D. Supplementary QR menu and ordering. Show dish details, availability, completed-order popularity or buffet take-up trends and estimated waits. Support fast-casual/made-to-order orders, checking pending demand against physical stock.
Prototype Scope: Six months, one line, up to three positions and 4-6 registered dishes; excludes payments, delivery, procurement ERP and independent food-safety certification.
Expected Users & Roles: Staff monitor and refill trays; managers maintain menus/settings and review reports; customers view menus and submit or track applicable orders.

4. Technical Stack & Special Requirements

Instructions: Detail the hardware and software tools you intend to use. Highlight how your project incorporates contemporary technologies (AI, IoT, Cloud, etc.) and how it aligns with modern considerations like environmental-friendly design, bilingual/multi-language support, or cross-border deployability to meet Greater China business needs.

Development Technologies & Tools:

- **Hardware:** Raspberry Pi 5 with load cells and ADC interfaces (such as HX711 or NAU7802), an overhead camera, surface-temperature sensor, display and LEDs. Assess RGB-D after range and cost tests. Existing UNO R4 WiFi is optional for acquisition/debugging.
- **Programming Languages:** Python, TypeScript, HTML/CSS; C++ only if a microcontroller is used.
- **Frameworks / Libraries:** FastAPI, OpenCV, PyTorch for model training, scikit-learn, React and i18next.
- **Database / Storage:** PostgreSQL and local measurement/image logs.
- **Target Platforms:** Local Linux device and responsive staff/customer web interfaces.

Special Project Considerations:

- **Contemporary Technologies:** IoT sensing, local AI inference and live web updates; optional cloud backup after the local core works.
- **Environmental-Friendly Solution:** Evaluate avoidable food surplus; reuse available computing equipment and use replaceable sensing components.
- **Bilingual / Multi-Language Support:** English and Traditional Chinese interfaces.
- **Cross-Border Deployment / Greater China Needs:** Configurable language, currency and time zone, with local deployment options for later regional adaptation.
- **Feasibility:** Confirm camera/ADC compatibility, sampling stability and cost by bench tests. Use controlled service scenarios initially and arrange a field trial if access becomes available.

5. Endorsements & Supervisor Feedback

Student Group Endorsement	By submitting this suggestion, we confirm that all listed group members are in agreement regarding this project concept and our assigned roles. Student Representative Signature: _____ Date: _____
Supervisor Vetting Decision Supervisor, please put a ☑ in the appropriate box.	☐ **APPROVED — The group is authorized to proceed to draft their formal Project Proposal.** ☐ **REJECTED — The group must modify their project topic suggestion and resubmit for review.**
Supervisor Feedback / Comments	
Supervisor Endorsement	Project Supervisor Signature: _____ Date: _____